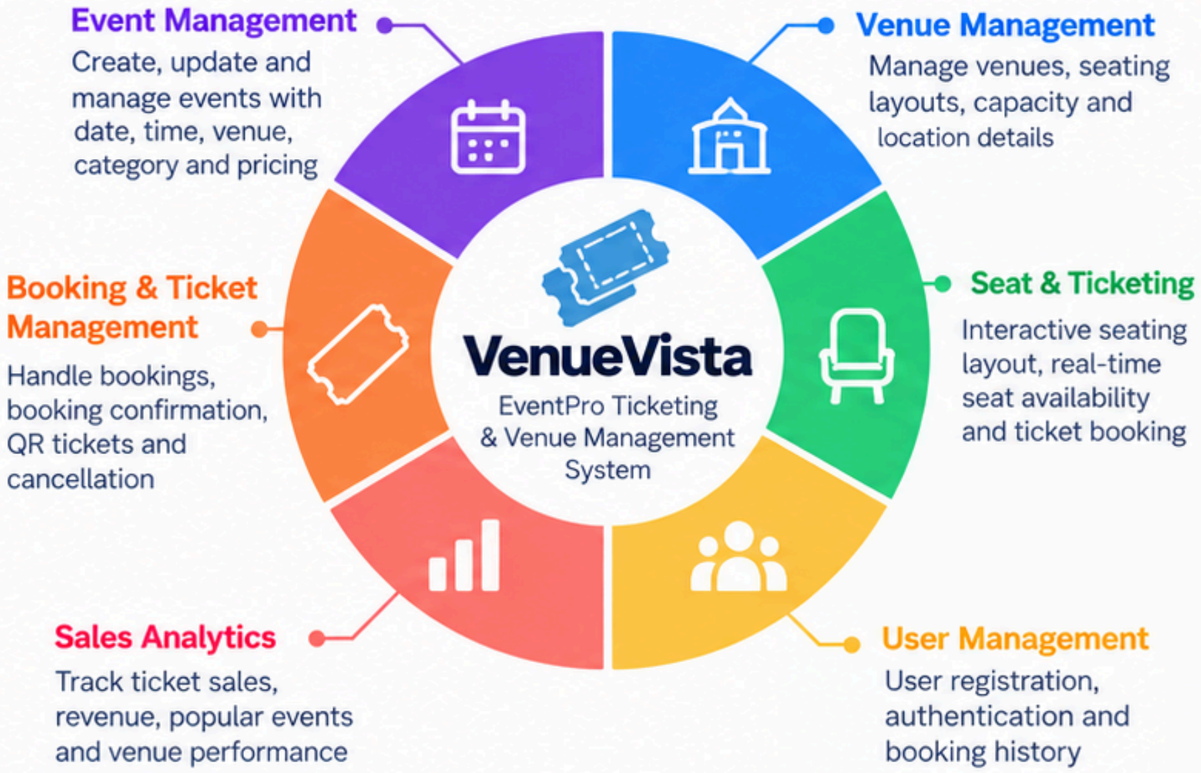
VenueVista

EventPro Ticketing & Venue Management System

Discover Events | Manage Venues | Book Tickets | Track Sales

Our Goal

To provide a simple and reliable platform for managing venues and events, enabling users to easily discover, book tickets and giving organizers powerful tools to manage operations and sales.



How it works

1. **Register / Login**
   Users and organizers create an account and login to the system
2. **Create Venue & Event**
   Admin/Organizer adds venues, configures seating and creates events
3. **Browse & Book**
   Users browse events, select seats and book tickets
4. **Confirm Booking**
   System generates booking details and e-tickets (QR Code)
5. **Manage & Analyze**
   Admins track bookings, sales and view reports for better decision making

Key Features

Event Management
Create, edit and manage events with category, date, time and pricing

Venue Management
Manage multiple venues with seating layout, capacity and location

Interactive Seat Layout
View real-time seat availability and select seats while booking

Ticket Booking
Bock tickets with a simple and seamless flow

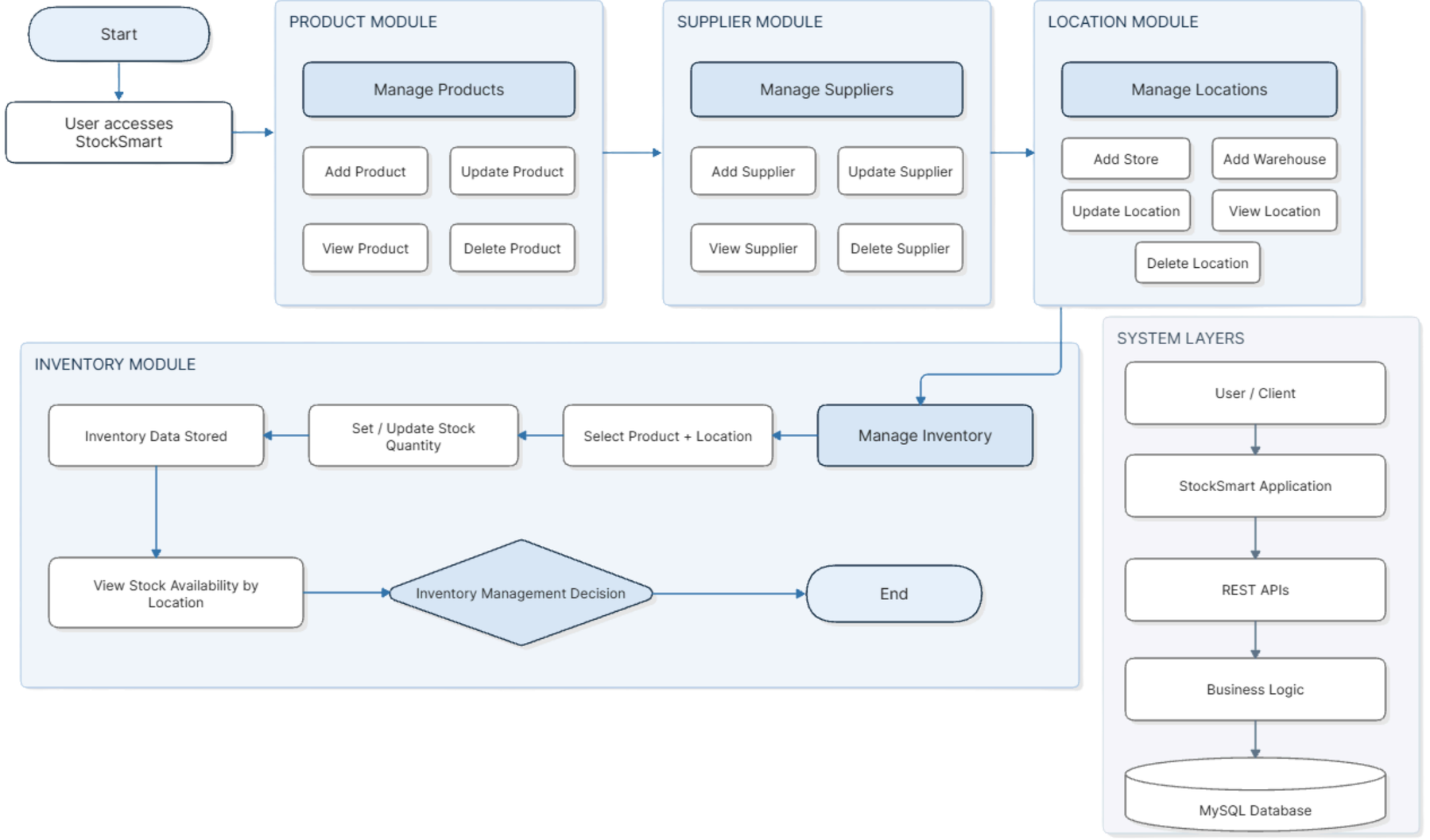
E-Tickets
Generate digital tickets with QR code for easy verification

User Management
Registration, login, profile management and booking history

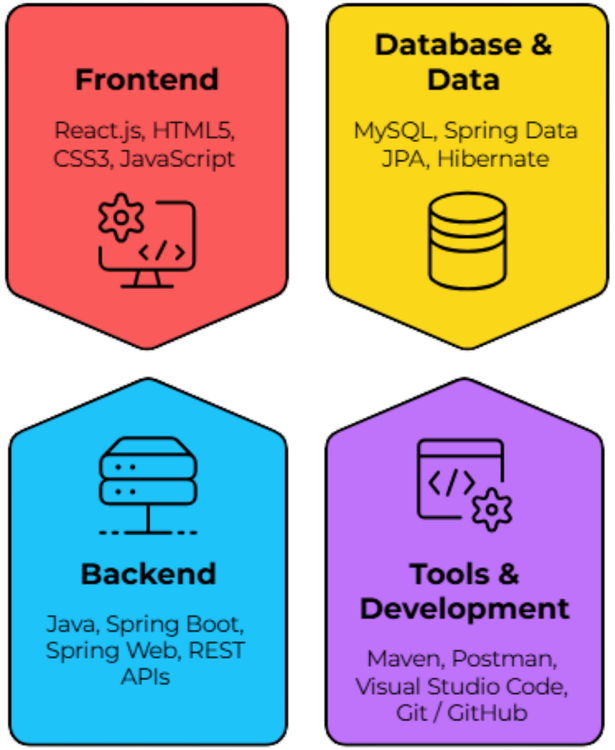
Sales Analytics
View total bookings, revenue, event-wise and venue-wise reports

Secure & Role Based
Separate access for Users and Admins with authentication and authorization

TECHNICAL APPROACH



Tech Stack Used



FEASIBILITY AND VIABILITY

Building a Practical and Scalable Event Ticketing & Venue Management System

Feasibility of VenueVista

The project is practical, cost-effective and can be implemented using current technologies to provide a complete solution for event booking and venue management.

Technical Feasibility

Can be developed using Java, Spring Boot, Angular and Oracle DB. Uses standard and widely available technologies and frameworks.

Operational Feasibility

Simple and user-friendly system for event organizers, venue managers and attendees. Requires basic training and can be easily operated.

Economic Feasibility

Cost-effective solution using open-source technologies. Reduces manual work, improves ticket sales and venue utilization, providing better ROI.

Legal Feasibility

Complies with data privacy regulations. Handles user data securely with proper authentication and authorization.

Schedule Feasibility

Can be developed within the planned timeline with the available resources and team size.

Scalability Feasibility

Designed in a modular way to support more venues, events, users and advanced features in the future without major changes.

Potential Challenges and Risks

High Traffic During Events

Sudden spike in users may cause performance issues during popular events.

Seat Overbooking

Incorrect seat locking or concurrent bookings can lead to double booking issues.

Data Security Risks

Unauthorized access or data breaches can compromise user information and bookings.

Venue Data Accuracy

Incorrect venue capacity or seating layout may lead to booking and management issues.

Integration & Future Expansion

Adding payment gateways, real-time notifications or other advanced features may require additional development effort.

Strategies for Overcoming These Challenges

Scalable Architecture

Use modular design, load balancing and optimized queries to handle high user traffic.

Seat Locking Mechanism

Implement real-time seat locking with database transactions to prevent double booking.

Secure Authentication & Authorization

Use JWT, role-based access and data encryption to protect user and booking data.

Regular Backup & Monitoring

Use scheduled backups, monitoring tools and proper logging to ensure system reliability.

Phased Feature Development

Start with core features and add advanced features (payments, notifications, analytics) in phases based on user feedback.

IMPACT AND BENEFITS

Creating a Better Event Experience for Everyone

Potential Impact on the Target Users

Event Attendees
Easily discover events, select seats, and book tickets with a simple and seamless experience.

Event Organizers
Manage events, venues, ticket pricing and track bookings in one place.

Venue Managers
Manage venue details, seating layout, capacity and availability efficiently.

Administrators
Monitor bookings, sales, users and venues through a centralized dashboard.

Local Businesses
Gain more audience and revenue through organized event management.



Visitors / Tourists
Find local events easily in a new city and book tickets to explore guided experiences.

Authoritis / Event Hosts
Better crowd management, nrganized data and improved safety through structured event information.

Students & Communities
Get access to college fests, cultural events, workshops and sports events easily.

Benefits of VenueVista

Increased Event Reach
Helps organizers reach a larger audience through an online platform.

Efficient Management
Centralized management of events, venues, seats and bookings reduces manual work and errors.

Higher Revenue Opportunities
Better event visibility, easy booking and analytics help increase ticket sales and revenue.

Improved Safety and Organization
Structured attendee data and digital tickets help in better crowd management for large events.

Better User Experience
Simple interface for searching events, selecting seats and booking tickets with real-time availability.



Scalability
Modular architecture supports future features like online payments, notifications and multi-city events.

Data-Driven Decisions
Analytics and reports provide insights on ticket sales, popular events and venue performance.

RESEARCH AND REFERENCES



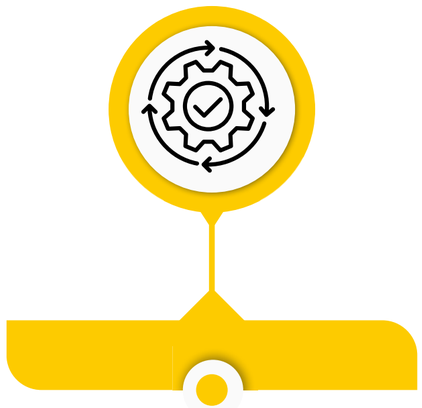
Study of online event discovery, booking, and digital ticketing workflows.



Research on venue capacity, seating layouts, and event scheduling.



Analysis of real-time seat availability and duplicate-booking prevention.



Reference for building secure REST-based full-stack applications.



Research on relational data management, entity relationships, and transaction handling.